

Normalization

1NF: Atomic values

Violation: An early draft of the **Municipality** table contained a **housing_types** column holding comma-separated lists. This violates 1NF because cells must hold single, indivisible values.

municipality_id	municipality_name	housing_types
1	Maastricht	Social Rental, Private Rental

After: Each cell contains exactly one value

municipality_id	municipality_name	housing_type_name
1	Maastricht	Social Rental
1	Maastricht	Private Rental

2NF: No Partial Dependencies

Violation: We drafted a combined statistics table with a composite Primary Key (**municipality_id** + **housing_type_id**). We included **municipality_name** here, but it only depends on the **municipality_id**, not the full composite key.

Why it's bad: If a municipality changes its legal name, we must update it in hundreds of rows across every recorded year and housing type, risking data inconsistency.

municipality_id (PK)	housing_type_id (PK)	municipality_name	average_rent
1	10	Maastricht	800.00
1	20	Maastricht	1000.00

After: We extracted `municipality_name` into its own `Municipality` entity and `housing_type_name` into the `Housing Type` entity.

municipality_id (PK)	municipality_name
1	Maastricht

municipality_id (PK)	housing_type_id (PK)	average_rent
1	20	1000.00

3NF: No Transitive Dependencies

Violation: Our `Municipality` table included both `province_id` and `province_name`. Here, `province_name` depends entirely on `province_id` (a non-key attribute), rather than directly on the `municipality_id` (the Primary Key).

Why it's bad: This causes massive data duplication; every single municipality in Limburg will repeat the text "Limburg" unnecessarily.

municipality_id (PK)	municipality_name	province_id	province_name
1	Heerlen	5	Limburg
2	Maastricht	5	Limburg

After: We created a distinct `Province` entity, leaving only the `province_id` as a Foreign Key in the `Municipality` table.

province_id	province_name
5	Limburg

municipality_id (PK)	municipality_name	province_id
1	Heerlen	5

ERD:

